



Carl Zeiss Vision International GmbH
73430 Aalen

**Carl Zeiss Vision
International GmbH**

Turnstrasse 27
73430 Aalen
Germany
Phone: +49 (0)73 61 591-0
Fax: +49 (0)73 61 591-1312
Email: wendelin.niesl@zeiss.com

Division/Dept.: Business Enablement & IT

Date 10.04.2026

SRE Hands-On Assessment (Azure & Azure DevOps / GitHub)

We are pleased to present a hands-on challenge tailored for an **SRE**. This assessment assumes familiarity with **Azure cloud services, CI/CD pipelines, Infrastructure as Code, and observability practices**.

Submission Guidelines

You are required to submit your project or a link to your work within **7 days** of receiving this assessment to proceed with the next steps in our evaluation process.

Engineering Standards

You are encouraged to:

- Apply **industry-standard SRE and DevOps practices**
- Use **Infrastructure as Code (IaC)** for all cloud resources
- Implement **observability, reliability, and scalability mechanisms**
- Follow **clean architecture and automation principles**

You are welcome to extend the solution with additional features that improve reliability, maintainability, or cost efficiency.

Sitz der Gesellschaft:
73430 Aalen
Deutschland
Handelsregister:
Ulm, HRB 501573
USt.-IdNr.: DE 237 102 714

Deutsche Bank AG Heidenheim
Konto 2 100 253 00 (BLZ 613 700 86)
S.W.I.F.T.-address: DEUT DESS 613
IBAN DE31 6137 0086 0210 0253 0

Geschäftsführung:
Sven Hermann (Vorsitzender)
Paul Bildsdörfer
Internet: www.zeiss.com/vision
E-Mail: info.vision@zeiss.com

Project Overview

Your task is to build a **cloud-native service platform on Azure**, including:

- A containerized API service
- Background processing via messaging
- CI/CD pipeline using Azure DevOps
- Infrastructure defined via code

Platform Requirements

Your solution should include:

- Compute:
 - Azure Container Apps (preferred) or alternative Azure compute
- Messaging:
 - Azure Storage Queue or Service Bus queue / topic
- Container Registry:
 - Azure Container Registry / Docker Registry
- Observability:
 - Azure Monitor with Application Insights
- CI/CD:
 - Azure DevOps pipelines or GitHub Actions
- Secrets:
 - Azure Key Vault

API Requirements

Expose a simple API (technology of your choice, preferably .NET):

Endpoints

- GET /api/work
- POST /api/work

Functional Behavior

- POST /api/work:
 - Enqueue a message to Service Bus
- Background worker:
 - Processes messages asynchronously
- GET /api/work:
 - Returns processed items (in-memory or persisted)

CI/CD Requirements

Implement a pipeline in Azure DevOps / GitHub Actions that:

Build Stage

- Builds the application
- Builds and tags a container image
- Pushes image to ACR / Docker Registry

Deploy Stage

- Deploys infrastructure (IaC)
- Deploys application to Azure Container Apps
- Uses environment-based configuration (dev/prod)

Infrastructure Requirements (IaC)

Use **Bicep or Terraform** to provision:

- Container Apps environment
- Service Bus namespace + queue
- ACR
- Application Insights

Security Requirements

- No secrets in code or pipeline YAML
- Use:
 - Managed Identity OR
 - Azure Key Vault integration

✅ Self-Check List Before Submission

- ☐ Ensure that your solution is **fully deployable via pipeline**
- ☐ Apply best practices for **CI/CD, IaC, and observability**
- ☐ Implement all required API endpoints
- ☐ Include **health probes (liveness + readiness)**
- ☐ Ensure **secure handling of secrets (no hardcoding)**
- ☐ Provide clear **README with setup and architecture explanation**
- ☐ Ensure infrastructure is reproducible via IaC

Deliverables

- Source code repository
- Infrastructure code (Bicep/Terraform)
- Azure DevOps pipeline / GitHub Actions YAML
- README including:
 - Architecture overview
 - Deployment steps
 - Design decisions
 - Known limitations

We look forward to reviewing your submission!

Best Regards,

Carl Zeiss Vision International
Business Enablement & IT Team